

AI ASSISTED LAB ASSIGNMENT-6.1

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BATCH:14

Task-1: (AI-Based Code Completion for Loops)

Prompt:

“Generate Python code to print all even numbers between 1 and N

Using a loop.”

Input:

#Generate Python code to print all even numbers between 1 and N using a loop.

```
N = int(input("Enter a number: "))  
print("Even numbers between 1 and", N, "are:")  
for i in range(1, N + 1):
```

```
if i % 2 == 0:
```

```
    print(i)
```

output:

```
PS C:\Users\Asus\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Asus\AppData\Local\Programs\Python\Python314\python.exe c:/Users/Asus/Untitled-2.py
Enter the value of N: 4
Even numbers between 1 and 4 are:
2 4
PS C:\Users\Asus\AppData\Local\Programs\Microsoft VS Code> |
```

Task-2: (AI-Based Code Completion for Loop withConditionals)

Prompt:

“Generate Python code to count how many numbers in a list are even and odd.”

Input:

```
def count_even_odd(numbers):
```

```
    even_count = 0
```

```
    odd_count = 0
```

```
    for num in numbers:
```

```
        if num % 2 == 0:
```

```
            even_count += 1
```

```
        else:
```

```
            odd_count += 1
```

```
    return even_count, odd_count
```

```
# Example usage
```

```
numbers_list = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
```

```
even_count, odd_count = count_even_odd(numbers_list)
print(f"Even numbers count: {even_count}")
print(f"Odd numbers count: {odd_count}")
```

output:

```
PS C:\Users\Asus\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Asus\AppData\Local\Programs\Python\Python314\python.exe c:/Users/Asus/Untitled-2.py
Even numbers count: 4
Odd numbers count: 3
PS C:\Users\Asus\AppData\Local\Programs\Microsoft VS Code> █
```

Task-3: (AI-Based Code Completion for Class Attributes Validation)

Prompt:

“Generate a Python class User that validates age and email using conditional statements.”

Input:

```
class User:
    def __init__(self, name, age, email):
        self.name = name
        self.age = age
        self.email = email

    def validate_age(self):
        if self.age < 0:
            return "Age cannot be negative."
```

```

        elif self.age < 18:
            return "User is a minor."
        else:
            return "User is an adult."

def validate_email(self):
    if "@" not in self.email or "." not in self.email:
        return "Invalid email format."
    else:
        return "Email is valid."

# Example usage:
user1 = User("Alice", 25, "alice@example.com")
print(user1.validate_age()) # Output: User is an adult.
print(user1.validate_email()) # Output: Email is valid.

```

Output:

```

PS C:\Users\Asus\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Asus\AppData\Local\Programs\Python\Python314\python.exe c:/Users/Asus/Untitled-2.py
Test Case 1:
Age and email are valid

Test Case 2:
Invalid age

Test Case 3:
Invalid email

Test Case 4:
Invalid age
PS C:\Users\Asus\AppData\Local\Programs\Microsoft VS Code>

```

Task-4: (AI-Based Code Completion for Classes)

Prompt:

“Generate a Python class Student with attributes (name, roll number, marks) and methods to calculate total and average marks.”

Input:

```
class Student:

    def __init__(self, name, roll_number, marks):

        self.name = name

        self.roll_number = roll_number

        self.marks = marks


    def calculate_total(self):

        return sum(self.marks)


    def calculate_average(self):

        total = self.calculate_total()

        return total / len(self.marks) if self.marks else 0


# Example usage:

student1 = Student("Bob", 101, [85, 90, 78, 92])

print(f"Total marks for {student1.name}: {student1.calculate_total()}")

print(f"Average marks for {student1.name}: {student1.calculate_average()}")
```

output:

```
PS C:\Users\Asus\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Asus\AppData\Local\Programs\Python\Python314\python.exe c:/Users/Asus/Untitled-2.py
Student Name: Rahul
Roll Number: 101
Marks: [80, 75, 90, 85, 70]
Total Marks: 400
Average Marks: 80.0
PS C:\Users\Asus\AppData\Local\Programs\Microsoft VS Code> █
```

Task-5: (AI-Assisted Code Completion Review)

Prompt:

“Generate a Python program for a simple bank account system using class, loops, and conditional statements.”

Input:

#Generate a Python program for a simple bank account system using class, loops, and conditional statements.

class BankAccount:

```
def __init__(self, account_number, account_holder, balance=0):  
    self.account_number = account_number  
    self.account_holder = account_holder  
    self.balance = balance
```

```
def deposit(self, amount):
```

```
    if amount > 0:  
        self.balance += amount  
        print(f"Deposited: ${amount}. New balance: ${self.balance}.")  
    else:  
        print("Deposit amount must be positive.")
```

```
def withdraw(self, amount):
```

```
    if amount > 0:  
        if amount <= self.balance:  
            self.balance -= amount  
            print(f"Withdrew: ${amount}. New balance: ${self.balance}.")  
        else:  
            print("Insufficient funds.")  
    else:  
        print("Withdrawal amount must be positive.")
```

```
def check_balance(self):  
    print(f"Account Balance: ${self.balance}.")  
    #give example usage  
if __name__ == "__main__":  
    # Create a new bank account  
    account = BankAccount("12345", "John Doe", 1000)  
  
    # Check initial balance  
    account.check_balance()  
  
    # Deposit money  
    account.deposit(500)  
  
    # Withdraw money  
    account.withdraw(200)  
  
    # Check final balance  
    account.check_balance()
```

Output:

```
PS C:\Users\Asus\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Asus\AppData\Local\Programs\Python\Python314\python.exe c:/Users/Asus/Untitled-2.py
Enter account holder name: Anjali
```

```
--- Bank Account Menu ---
```

1. Deposit
2. Withdraw
3. Check Balance
4. Exit

```
Enter your choice: 1
```

```
Enter deposit amount: 5000
```

```
Amount deposited successfully.
```

```
--- Bank Account Menu ---
```

1. Deposit
2. Withdraw
3. Check Balance
4. Exit

```
Enter your choice: 2
```

```
Enter withdrawal amount: 1500
```

```
Amount withdrawn successfully.
```

```
--- Bank Account Menu ---
```

1. Deposit
2. Withdraw
3. Check Balance
4. Exit

```
Enter your choice: 3
```

```
Current Balance: 3500.0
```